

Course: CSN405 – Network Programming
LAB 3 – Socket Programming

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Question 1: Install the Python programs TCPClient and UDPClient on one host and TCPServer and UDPServer on another host.

- Suppose you run TCPServer and you try to connect using UDPClient. What happens? Why?
- Suppose you run UDPClient before you run UDPServer. What happens? Why?
- What happens if you hardwire in the python client and server programs different port numbers for the client and server sides in either a TCP or UDP client-server pair?

Question 2: Suppose that in UDPClient.py, after we create the socket, we add the line: `clientSocket.bind(('', 5432))`

- Will it become necessary to change UDPServer.py?
- What are the port numbers for the sockets in UDPClient and UDPServer?
- What were they before making this change?

Question 3: Write a simple client-server application using both TCP and UDP
Client Implementation

- Read a line of text from the keyboard.
- Send the text to the server.
- Receive the result from the server and display it.
- Allow the user to send multiple sentences (repeat until the user enters "exit").

Server Implementation

- Receive a line of text from the client.
- Count how many times the letter 'n' appears in the text.
- Send the count back to the client.

Note: remember using try / except / finally

Question 4: TCP True/False Quiz

Server Implementation

Create a TCP server that listens to localhost, port 5050.

Prepare a list of at least 3 True/False questions:

- “Port 443 is used for HTTP.”
- “TCP is a connection-oriented protocol.”
- “UDP guarantees delivery of packets.”
- “The send() method is used to receive data from a socket.”
- “The bind() method is used to connect a client to a server.”

When all questions are answered: send “Quiz finished!” to client

If the client sends quit or exit, the server should close the connection

When a client connects:

- Send the quiz questions to the client one by one
- For each question:
 - Receive the client’s answer (“true” or “false”)
 - Check whether the answer is correct
 - Send one of the following responses:
 - “Yes, it’s correct!” if the answer is correct
 - “No, it’s wrong!” if the answer is incorrect

Client Implementation

- Connect to the server on localhost, port 5050
- Receive each question from the server
- Prompt the user to enter “true” or “false”
- Send the answer to the server
- Display the server’s response
- Continue until all questions are finished or the user enters “quit” or “exit”